

List of Publications

- [1] E.D. Sontag. *Notes on Mathematical Systems Biology (online)*. Online only., 2026. Continuously updated. If the link does not work, then copy/paste this: <http://drive.google.com/drive/folders/1lIRqaCPeXMVZGoY-44bBsvtnsHtIRfIO?usp=sharing>.
- [2] E. D. Sontag. Dynamics and dose response in scaffold ligand binding. 2026. Submitted. Preprint in arXiv:2508.06599.
- [3] K. Manoj, D. Jatkar, M. Ali Al-Radhawi, E.D. Sontag, and D. Del Vecchio. Paradoxical gene regulation explained by competition for genomic sites. 2026. Submitted. Also bioRxiv 2025.11.27.691022.
- [4] M.K. Wafi, A.C.B de Oliveira, and E.D. Sontag. Boundedness of solutions in feedback systems with antithetic controllers. 2026. arXiv 2604.27290.
- [5] M.K. Wafi, A.C.B de Oliveira, and E.D. Sontag. When is cumulative dose response monotonic? analysis of incoherent feedforward motifs. In *Proc. 65th IEEE Conference on Decision and Control (CDC)*, 2026. Submitted. Also arXiv:2604.01573.
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- [7] A. C. B. de Oliveira, R. Wang, I.R. Manchester, and E. D. Sontag. Remarks on Lipschitz-minimal interpolation: Generalization bounds and neural network implementation. In *Proc. 65th IEEE Conference on Decision and Control (CDC)*, 2026. Submitted. Also arXiv:2603.19524.
- [8] E.D. Sontag. Dynamic response phenotypes and model discrimination in systems and synthetic biology. *arXiv 2512.24946*, 2025. Also in Authorea: <https://doi.org/10.22541/au.176790592.20368210/v1>.
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- [12] M. D. Kvalheim and E. D. Sontag. Autoencoding dynamics: Topological limitations and capabilities. *arXiv*, page 2511.04807, 2025. Journal version submitted.
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- [15] L. Cui, Z.P. Jiang, and E. D. Sontag. Small-covariance noise-to-state stability of stochastic systems and its applications to stochastic gradient dynamics. In *2026 American Control Conference (ACC)*, 2026. To appear. Also 2025 arXiv:2509.24277.
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- [17] E. D. Sontag. Dynamics of binding three independent ligands to a single scaffold. *arXiv*, page 2508.06599, 2025.
- [18] M.K. Wafi, A.C.B de Oliveira, and E.D. Sontag. On the (almost) global exponential convergence of overparameterized policy optimization for the LQR problem. In *2026 American Control Conference (ACC)*, 2026. To appear. See also 2025 arXiv:2510.02140.
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